

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Blu-ray
Manufacturer: Cambridge Audio
Firmware Version: N/A
Model(s): CXUHD

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
2.08.0002-b001	1.4.2

Version History

Module Version	Date	Notes
1_0_0_0	8/14/2018	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='CXUHD')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='CXUHD')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format without Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command Input	Value 'Blu-Ray Player' 'ARC HDMI Out 1' 'Coaxial'	Value 'Front HDMI/MHL' 'ARC HDMI Out 2'	Value 'Back HDMI' 'Optical'
# Input example InterfaceName.Set('Input', 'Blu-Ray Player')			
Command Keypad	Value '0' '3' '6' '9'	Value '1' '4' '7' 'Clear'	Value '2' '5' '8'
# Keypad example InterfaceName.Set('Keypad', '0')			
Command MenuNavigation	Value 'Home Menu' 'Top Menu' 'Down' 'Enter'	Value 'Page Up' 'Pop-up Menu' 'Left' 'Setup Menu'	Value 'Page Down' 'Up' 'Right' 'Return'
# MenuNavigation example InterfaceName.Set('MenuNavigation', 'Home Menu')			
Command Mute	Value None		
# Mute example InterfaceName.Set('Mute', None)			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command Repeat	Value 'Chapter' 'Shuffle'	Value 'Title' 'Random'	Value 'All' 'Off'
# Repeat example InterfaceName.Set('Repeat', 'Chapter')			

Global Scripter Module Communication Sheet

Command Transport	Value 'Stop' 'Previous' 'Fast Forward'	Value 'Play' 'Next' 'Eject Disc Tray'	Value 'Pause' 'Fast Reverse'
# Transport example InterfaceName.Set('Transport', 'Stop')			
Command TVSystemOutput	Value 'NTSC'	Value 'PAL'	Value 'Auto'
# TVSystemOutput example InterfaceName.Set('TVSystemOutput', 'NTSC')			
Command Volume	Value 0 to 100 in steps of 1		
# Volume example InterfaceName.Set('Volume', 100)			

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'}, FeedbackHandler)
FeedbackHandler will be called only when the specified qualifier gets a new status.
```

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command	Qualifier Value		
ChapterStatus	'String'		
# ChapterStatus examples InterfaceName.Update('ChapterStatus') Value = InterfaceName.ReadStatus('ChapterStatus') InterfaceName.SubscribeStatus('ChapterStatus', None, FeedbackHandler)			
Command	Value	Value	
ConnectionStatus	'Connected'	'Disconnected'	
# ConnectionStatus examples Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command	Value	Value	Value
Input	'Blu-Ray Player'	'Front HDMI/MHL'	'Back HDMI'
	'ARC HDMI Out 1'	'ARC HDMI Out 2'	'Optical'
	'Coaxial'		
# Input examples InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command	Value	Value	Value
PlaybackStatus	'No Disc'	'Loading'	'Open'
	'Close'	'Play'	'Pause'
	'Stop'	'Step'	'FREV'
	'FFWD'	'SFWD'	'SREV'
	'Setup'	'Home Menu'	'Media Center'
# PlaybackStatus examples InterfaceName.Update('PlaybackStatus') Value = InterfaceName.ReadStatus('PlaybackStatus')			

Global Scripter Module Communication Sheet

InterfaceName.SubscribeStatus('PlaybackStatus', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	
# Power examples InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command Repeat	Value 'Chapter'	Value 'Title'	Value 'All'
	'Shuffle'	'Random'	'Off'
# Repeat examples InterfaceName.Update('Repeat') Value = InterfaceName.ReadStatus('Repeat') InterfaceName.SubscribeStatus('Repeat', None, FeedbackHandler)			
Command Track/TitleStatus	Qualifier Value 'String'		
# TrackTitleStatus examples InterfaceName.Update('TrackTitleStatus') Value = InterfaceName.ReadStatus('TrackTitleStatus') InterfaceName.SubscribeStatus('TrackTitleStatus', None, FeedbackHandler)			
Command Volume	Value 0 to 100 in steps of 1		
# Volume examples InterfaceName.Update('Volume') Value = InterfaceName.ReadStatus('Volume') InterfaceName.SubscribeStatus('Volume', None, FeedbackHandler)			

Cable and Adapter Requirements

Captive Screw to Male DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 9600

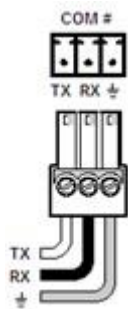
Data Bits: 8

Parity: None

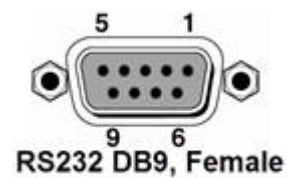
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD		2	TxD
RxD		3	RxD
GND		5	GND



Appendix A. Set Commands

Input ARC HDMI Out 1	#SIS 3\x0D
Input ARC HDMI Out 2	#SIS 4\x0D
Input Back HDMI	#SIS 2\x0D
Input Blu-Ray Player	#SIS 0\x0D
Input Coaxial	#SIS 6\x0D
Input Front HDMI/MHL	#SIS 1\x0D
Input Optical	#SIS 5\x0D
Keypad 0	#NU0\x0D
Keypad 9	#NU9\x0D
Keypad Clear	#CLR\x0D
Menu Navigation Down	#NDN\x0D
Menu Navigation Enter	#SEL\x0D
Menu Navigation Home Menu	#HOM\x0D
Menu Navigation Left	#NLT\x0D
Menu Navigation Page Down	#PDN\x0D
Menu Navigation Page Up	#PUP\x0D
Menu Navigation Pop-up Menu	#MNU\x0D
Menu Navigation Return	#RET\x0D
Menu Navigation Right	#NRT\x0D
Menu Navigation Setup Menu	#SET\x0D
Menu Navigation Top Menu	#TTL\x0D
Menu Navigation Up	#NUP\x0D
Mute None	#MUT\x0D
Power Off	#POF\x0D
Power On	#PON\x0D
Repeat All	#SRP ALL\x0D
Repeat Chapter	#SRP CH\x0D
Repeat Off	#SRP OFF\x0D
Repeat Random	#SRP RND\x0D
Repeat Shuffle	#SRP SHF\x0D
Repeat Title	#SRP TT\x0D
Transport Eject Disc Tray	#EJT\x0D
Transport Fast Forward	#FWD\x0D
Transport Fast Reverse	#REV\x0D
Transport Next	#NXT\x0D
Transport Pause	#PAU\x0D

Transport Play	#PLA\x0D
Transport Previous	#PRE\x0D
Transport Stop	#STP\x0D
TV System Output Auto	#SPN AUTO\x0D
TV System Output NTSC	#SPN NTSC\x0D
TV System Output PAL	#SPN PAL\x0D
Volume 0	#SVL 0\x0D
Volume 100	#SVL 100\x0D

Appendix B. Update Commands

Chapter Status	#QCH\x0D
Input	#QIS\x0D
Playback Status	#QPL\x0D
Power	#QPW\x0D
Repeat	#QRP\x0D
Track/Title Status	#QTK\x0D
Volume	#QVL\x0D